

Can a Cultural Code Be Computed: Extracting the Relational Grammar of Crimean Tatar Örnek

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Abstract. Crimean Tatar Örnek, inscribed in 2021 on UNESCO’s Representative List of the Intangible Cultural Heritage of Humanity, is not simply a repertoire of decorative motifs. It is a living symbolic system in which meaning emerges through selection, combination, enclosure, hierarchy, repetition, and spatial relationship. This paper asks whether the culturally specific logic -the cultural “code”- of Örnek can be made computationally legible without reducing it to visual style or transferring cultural authority to an algorithm. It proposes a provenance-aware relational model in which motifs are attributed nodes and culturally meaningful relations are typed edges. A controlled experiment compares random, geometry-only, semantic-relational, and full-grammar generation with a relational-ablation condition that preserves motif inventory and broad appearance while perturbing selected relations. A neighboring-tradition corpus scored against the same rules tests whether the model captures anything specific to Örnek. The contribution is not an automated claim to authenticity, but an inspectable, community-correctable instrument for studying how inherited visual knowledge can remain active within computational media.

Keywords: Crimean Tatar Örnek, cultural code, relational grammar, computational heritage, intangible cultural heritage, procedural generation

1. Introduction: The Past Meets the Future

Digital heritage systems preserve visible form with increasing precision. Photography, vectorization, photogrammetry, and generative models can reproduce color, geometry, texture, and stylistic appearance. Intangible cultural heritage, however, also consists of practices, representations, knowledge, and skills that communities recognize and continually recreate [1]. Digital circulation can aid safeguarding while raising questions of local cultural property rights and of who controls documentation and reuse [2]. The challenge is therefore not only to digitize what the past looked like, but to understand how its visual systems produced and transmitted meaning.

This problem is especially clear in Crimean Tatar Örnek. UNESCO describes Örnek as a system of symbols used across embroidery, weaving, pottery, engraving, jewelry, wood carving, glass, and wall painting. Around thirty five symbols participate in narrative compositions understood and commissioned by community members [3]. Public documentation associates the rose with a married woman, the tulip with a young man, the almond with an unmarried woman or girl, and the carnation with age, wisdom, and experience. Motifs do not communicate only in isolation: a tulip within a rose signifies the love or union of a man and a woman [3]. The relationship changes the message.

This paper uses the phrase cultural or “national code” cautiously. It does not mean an immutable national essence, but a historically transmitted and continuously interpreted system combining motif vocabulary, semantic convention, spatial relationship, material practice, provenance, and practitioner knowledge. The central question is whether computation can reveal this relational structure while allowing the living tradition- and its knowledge holders -to remain the source of meaning.

2. From Visual Style to Relational Cultural Syntax

A model can reproduce Örnek-like palettes, curves, symmetry, and repetition while omitting the relations through which a composition communicates. Visual resemblance is therefore an insufficient test. I propose *relational cultural syntax* as a computational hypothesis: Örnek’s communicative identity may arise through constrained networks of enclosure, attachment, pairing, hierarchy, direction, repetition, and compositional position.

This hypothesis is defined against an existing formal tradition. Generative accounts of ornament descend from shape grammars [4], which specify designs through geometric rewriting rules and have been surveyed across five

decades of engineering and architectural application [5]. Parametric shape grammars have generated Moorish geometric designs [6], and recent work combines grammars with neural networks or compares them with diffusion models for traditional pattern innovation [7, 8]. Such systems formalize how form is produced; they do not necessarily represent what a configuration means to the community that reads it. A shape grammar can derive a tulip inside a rose without encoding that the enclosure denotes union. The model proposed here is therefore complementary: typed semantic relations sit above a geometric layer that a shape grammar could supply, and it is the semantic layer, not the geometric one, that the experiment manipulates.

This reframing also offers a method for investigating what may distinguish Örnek from neighboring traditions. Flowers, trees, vines, and geometric forms circulate widely across Islamic, Ottoman, and regional decorative arts; uniqueness cannot be declared from isolated shapes. Comparison must ask how motifs are combined, nested, ranked, oriented, and connected, and whether these relationships support culturally recognized narratives. Örnek’s distinctive “code” may lie less in exclusive motifs than in the organization of a shared visual vocabulary through particular semantic conventions. This claim remains comparative and testable. A corpus should include documented historical and contemporary Örnek across media together with carefully selected neighboring material, each item carrying object-level provenance: maker where known, date, region, medium, collection, documented interpretation, and permission status. Recurrence may suggest a rule, but frequency alone cannot turn a pattern into cultural knowledge.

3. A Provenance-Aware Computational Model

A composition is represented in three coupled layers. The geometric layer records approximate symmetry, density, overlap, hierarchy, curvature, and negative space. The relational layer represents a composition as an attributed directed multigraph $C = (V, E)$: vertices are motif instances with identity, semantic role, scale, orientation, position, and provenance; edges encode relations such as enclosed-by, attached-to, paired-with, mirrors, repeats, grows-from, or subordinate-to. Multiple edges may connect the same motifs because spatial and semantic relations can coexist.

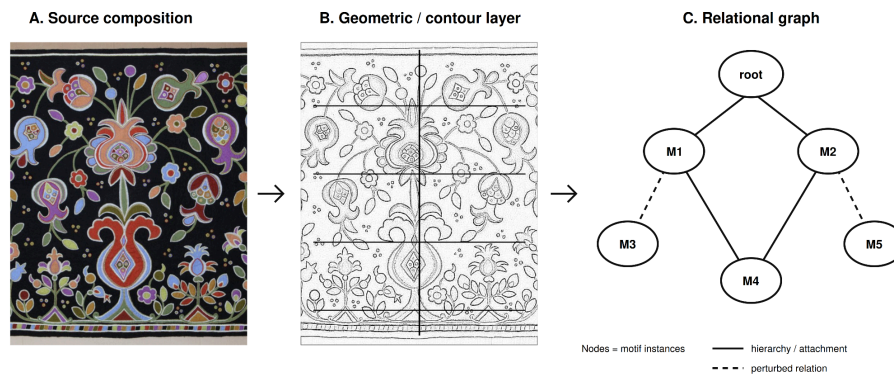


Figure 1: Original source-to-model graphic: source composition, contour abstraction, and an illustrative relational graph. BlackSeaNews (2018)

The semantic layer combines motif-level meanings with meanings produced by relationships. Figure 1 illustrates the analytical pipeline from a source composition to geometric description and a provisional relation graph; it does not claim that the schematic node labels identify the depicted motifs. Rules receive an explicit epistemic status - community-validated, documented, corpus-supported, or exploratory - so users can see how each proposition entered the model. Encoded-rule conformity is the proportion of typed edges licensed by the current knowledge base, reported separately by epistemic tier. It is described as conformity, never cultural validity: it records consistency with a particular, revisable knowledge base and certifies nothing about authenticity.

Generation reverses the analysis. A semantic-relational grammar first selects an intended narrative, motifs, and supported relationships. A geometric grammar then realizes that graph spatially. Every output retains a machine readable record containing its seed, motif instances, graph edges, intended meaning, violated

constraints, provenance, permissions, epistemic status, and software version. The image is therefore evidence of a process, not the sole research object.

4. Experimental Test: Does Relationship Carry Meaning?

The core experiment compares five conditions, listed here in the left-to-right order of Figure 2: (A) random placement, (B) geometry-only generation, (C) relational ablation, (D) semantic-relational generation, and (E) the full grammar combining semantic relationships with geometric realization. This is a display order rather than a processing order: ablation is derived from an already generated full-grammar composition, so (C) presupposes (E). Motif inventory and seeds are matched across conditions, so differences cannot be attributed to the number or identity of motifs. Reference and generated compositions are compared through geometry, graph structure, motif co-occurrence, recurring subgraphs, hierarchy, branching, and encoded-rule conformity.

Relational ablation (C) provides the strongest test. Starting from a documented or full-grammar composition, it preserves motif inventory and, as far as possible, broad symmetry, density, and spacing while perturbing one selected relation per composition, such as enclosure, attachment, pairing, or hierarchy. Two controls guard against a trivial result. First, a manipulation check confirms that intact and ablated compositions are statistically indistinguishable on the geometric feature vector at a pre-specified threshold, so reduced interpretability cannot be attributed to visible disturbance alone. Second, participants are blinded to condition, and practitioners who supply relation annotations are distinct from those who evaluate outputs. If practitioners find the altered composition less interpretable even though its motifs and broad appearance remain similar, this supports the hypothesis that culturally meaningful information resides partly in relational structure.

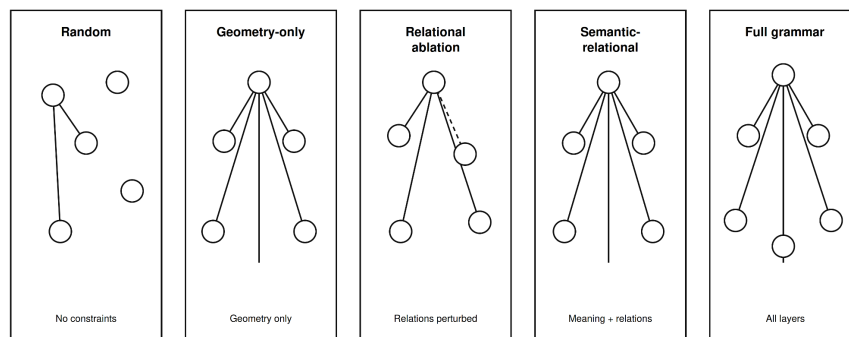


Figure 2: Original controlled-generator graphic. Conditions run left to right as random (A), geometry-only (B), relational ablation (C), semantic-relational (D), and full grammar (E). The panels are schematic: node counts and layouts illustrate the constraint regime of each condition rather than the matched motif inventory used in the experiment itself, in which inventory and seeds are held constant. Only C perturbs a relation while retaining broad geometry.

A comparative condition scores a matched corpus of neighboring Ottoman and wider Islamic ornament against the same encoded rules. This makes the distinctiveness claim falsifiable: if Örnek and neighboring material achieve similar conformity, the rule set captures shared ornamental convention rather than anything specific to Örnek. Scale and protocol determine whether the result is interpretable. The reference corpus targets about two hundred documented compositions across at least five media, each annotated independently by two annotators; agreement on relation type is reported with Krippendorff's α , and low-agreement relations remain exploratory. Evaluation requires twelve to twenty Crimean Tatar artists, craftspeople, researchers, and community knowledge holders. Participants assess whether a composition is interpretable, whether an intended message is plausible, which relations are ambiguous or invalid, and whether the ontology distorts living practice. Open explanations are more important than collapsing responses into a single "authenticity" score. Disagreement is retained because it may reflect regional, generational, material, or individual variation rather than error.

5. Ethics, Limits, and Community Authority

Formalization can preserve knowledge, but it can also freeze flexible meanings, enable unattributed commercial reuse, and make numerical scores appear culturally authoritative. The system must therefore store

provenance and permissions at the level of objects, motifs, and rules; support restricted or removed examples; document uncertainty and disagreement; and never label generated work as traditionally authentic.

The 1944 deportation and the continuing occupation of Crimea shape access to objects, archives, and practitioners; knowledge holders are dispersed across Crimea, mainland Ukraine, Turkey, and a wider diaspora [10, 11]. Corpus coverage will therefore reflect displacement as much as history, and the model must record that unevenness rather than smooth it. Governance adopts the CARE Principles - collective benefit, authority to control, responsibility, and ethics [9] - so that annotations and rule sets remain revocable by the communities they describe. The ontology also builds on the practitioner scholarship through which Örnek was systematized and revived, including the work of Mamut Churlu and the craft networks documented in the nomination file [10], rather than on the inscription summary alone.

This study is a framework and experimental protocol, not a completed validation. Public descriptions of motif meanings are incomplete; no proposed corpus can prove national uniqueness; and historical and contemporary Örnek contain regional, material, and individual variation. Comparison must distinguish observation from interpretation and avoid treating a computational ontology as the tradition itself. Community consultation is not a final label applied after engineering: it is part of knowledge production and may revise the ontology, invalidate rules, restrict examples, or change the research question.

6. Conclusion

At the meeting point of past and future, the essential question is not whether AI can imitate Örnek, but whether computation can help expose the relationships through which Örnek carries memory, identity, and knowledge. The proposed framework treats motifs as more than visual tokens and generation as more than stylistic synthesis. Its relational-ablation experiment makes the central claim falsifiable: if interpretability survives only while relationships remain intact, relational organization is part of the cultural information being transmitted.

The intended future is not a machine that replaces interpretation, but an inspectable and community-correctable research instrument through which inherited cultural knowledge can be compared, questioned, and transmitted without concealing provenance or uncertainty. Computation becomes valuable not when it claims authority over the past, but when it creates new ways for the past to remain legible, revisable, and alive.

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